

■ CYCLONE SENTINEL

AI-Powered Disaster Intelligence & Infrastructure Protection Platform

Build with AI: Code for Communities Hackathon Submission

Prototype: <https://cyclone-sentinel-delta.vercel.app>

GitHub: <https://github.com/Saicharan-bit1616/cyclone-sentinel>

The Challenge: Coastal Disaster Response Gaps

Why Traditional Emergency Operation Systems Fail During Extreme Storm Events

1. Fragmented Hazard Telemetry

Meteorological storm track data is isolated from local municipal infrastructure maps, causing delayed emergency responses during rapid landfall changes.

2. Critical Facility Vulnerabilities

Hospitals, 220kV power substations, and water networks face cascading failures (blackouts, submergence) without pre-landfall protective staging.

3. Manual Operational Bottlenecks

Emergency operations centers waste vital pre-landfall hours manually auditing high-risk facilities instead of executing automated triage.

4. Offline Reliability Failure

Traditional cloud tools crash during storm-induced internet outages, leaving decision-makers blind when real-time intelligence is needed most.

The Solution: Cyclone Sentinel Architecture

AI-Driven Operational Decision-Support for Emergency Command Centers

■ Interactive Geospatial Satellite Map

Displays live storm track corridors, wind intensity rings, coastal flood polygons, and color-coded infrastructure facility markers with Leaflet GIS visualizers.

■ Gemini 2.5 Flash AI Triage Engine

Dynamically evaluates asset risk ratings, elevation, served population, and structural vulnerability drivers to generate 3-step actionable mitigation steps.

■ Sector Infrastructure Matrix

Categorized monitoring across Hospitals, Power Substations, Evacuation Shelters, Potable Water Plants, and Primary Transport Corridors.

■ Zero-Downtime Rule Engine Fallback

Deterministic offline fallback engine guarantees continuous operational triage even during network disconnection or API rate limits.

Technology Stack & Architecture

Modern, Resilient Web Architecture Built for Speed and High Availability

Component	Technology	Purpose & Advantage
Frontend Core	React 19 + TypeScript + Vite	Ultra-fast rendering, type-safe asset modeling, instant HMR.
UI & Styling	Tailwind CSS + Lucide Icons	Modern Glassmorphism emergency command dashboard layout.
Geospatial GIS	Leaflet + React-Leaflet	Custom tile layers, hazard polygons, and dynamic asset markers.
AI Reasoning	Google Gemini AI (@google/genai)	Structured JSON tactical triage via Gemini 2.5 Flash API.
Fallback Triage	Deterministic TypeScript Engine	Zero-downtime offline rule-based operational fallback.
Deployment	Vercel Edge Network	Global CDN edge hosting with instant CI/CD deployment.

Community Impact & Scalability

Protecting Vulnerable Populations & Mitigating Multi-Million Dollar Infrastructure Losses

♥■ Life Safety & Triage

Ensures intensive care units & shelters have pre-positioned 500kVA generators and clear 4x4 ambulance corridors before storm surge landfall.

■ Power & Utility Protection

Protects 220kV switchyards with sandbag perimeters and dewatering pumps, preventing blackout cascades for 1.2M+ residents.

■ Global Adaptability

Designed to ingest any coastal weather feed or municipal asset dataset, making it deployable for hurricane and typhoon zones worldwide.

■ Hackathon Deliverable Status

100% functional prototype deployed live on Vercel with GitHub source code publicly available under open license.